Foreign Banks, Exporters, and Global Financial Exposure

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Fall 2026

This project received data access and support from the EconLab at Banco de Mexico.
The views and opinions expressed do not reflect the positions of Banco de Mexico.

Motivation

- International banks are central to global finance, yet countries differ sharply in how far they open their banking systems to them cross-country
- **Q:** What are the benefits and costs of hosting foreign banks?
  - Foreign banks characterized as more efficient, conduits of cheap funding and promoters of trade (Niepmann '23, Claessens-van Horen '21, Caballero et al. '18)
  - Foreign banks rely on fickle foreign funding, amplifying the transmission of global financial shocks (Cetorelli-Goldberg '11, Schnabl '12, Morais et al. '19, Aldasoro et al. '22)
- We study both sides of the argument by bringing micro-data from Mexico to a macro-finance model of bank–firm interactions:
  - Bank balance sheets and a bank–firm credit registry matched to customs data
  - A quantitative small-open-economy model with banks that differ in funding and ownership, firm–bank sorting, and cross-border borrowing

Preview of Results

What do foreign banks do? How do they affect the economy that hosts them?

- **Facts:** Foreign banks disproportionately lend to exporters, fund themselves abroad, and cut credit by more than domestic banks when US rates rise
- **Model:** Banks that differ in funding and borrowers. Foreign banks tighten more because their funding cost tracks world rates; lending to exporters affects how the shock reaches the economy
- **Quantifying** the benefits and costs of hosting foreign banks:
  - **Level gains:** cheaper credit matters little; the gain comes from trade services to exporters, up to 2% of steady-state consumption at the upper end of our estimates
  - **Amplifying exposure to foreign shocks:** a modest 4–9% larger response of consumption to a foreign interest rate shock
  - **Regulation mitigates the trade-off:** an existing cap on banks' net dollar position keeps the exposure small at a steady-state cost of 0.4% of consumption

Selected Literature

Global transmission of shocks, US MP spillovers, GFCy: **Bruno and Shin '15, '23**; **Morais et al. '19**; **Aoki et al. '21**; di Giovanni et al. '21; Miranda-Agrippino and Rey '20, '22; Obstfeld and Zhou '23; Akinci et al. '24; Camara et al. '25

- Study the transmission of financial and demand shocks within a common macro-finance model

Financial openness & international banks: Crystal, Dages, Goldberg '01; Moreno and Villar '05; Gourinchas and Jeanne '06; Detragiache et al '08; Cetorelli and Goldberg '11, '12; Caballero et al '18; Aldasoro et al '22; Niepmann '23

- Evidence and model of international banks as both trade facilitators and shock transmitters

Trade and bank finance: Amiti and Weinstein '11; Chor and Manova '11; Minetti and Zhu '11; **Paravisini et al. '15, '23**; Niepmann and Schmidt-Eisenlohr '17; Claessens and Van Horen '21; Xu '22

- Explain how bank lending to exporters connects trade outcomes and financial conditions

Real effects of credit supply shocks: Khwaja and Mian '08; Jiménez et al. '12, '20; Chodorow-Reich '14; Amiti and Weinstein '18; **Herreño '25**; **Rivadeneira et al. '25**

- Map micro-level credit-supply effects into aggregate outcomes in general equilibrium

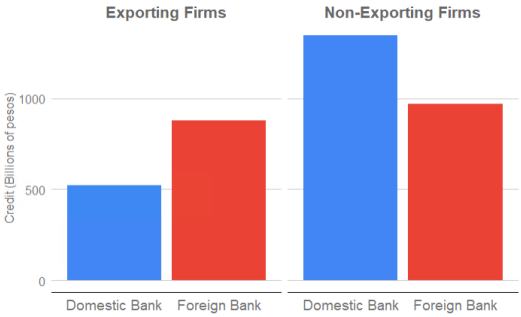
Setting, Data & Descriptive Facts

Mexico: Context and Data

- 50 banks, $\approx$ 20 of which are *local offices of foreign-owned banks* details
- Exports are $\approx$ 40% of GDP. 90% manufacturing. $>$ 85% to US details

- Credit loan-level data:
  - Monthly reports from 2003 to 2024 with matched bank–firm data for the universe of loans
  - $\approx$ 1 million credits to 350k firms each period. Focus on active loans from commercial banks
- Customs firm-level data:
  - Monthly data since 2006 with the dollar value of firm exports and imports
  - Around 35000 firms export every year. We call any firm that ever exports an exporter
- Supervisory bank balance-sheet data, including cross-border liabilities

Foreign Banks Disproportionately Lend to Exporters



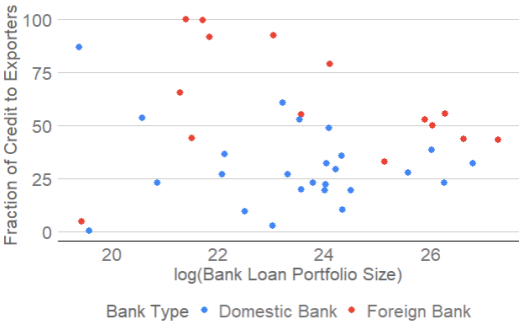
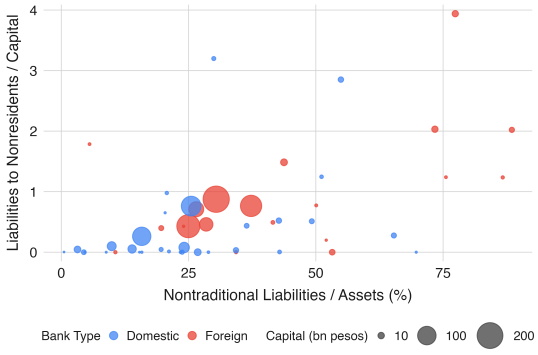


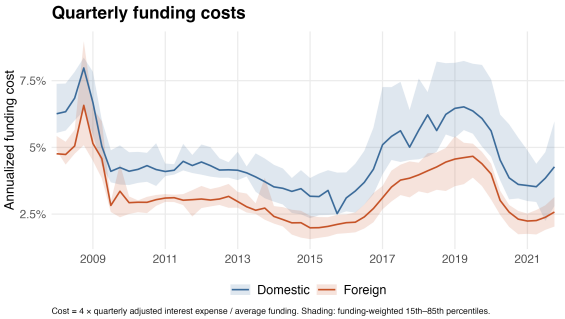
Figure: Outstanding loan balances (2023)

- Exporters get 20 pp more of their credit from international banks
- Persistent, not driven by FX loans, holds in firm- and bank-level reg. with controls

▸ loan pricing

Foreign Banks Fund Themselves Cross-Border

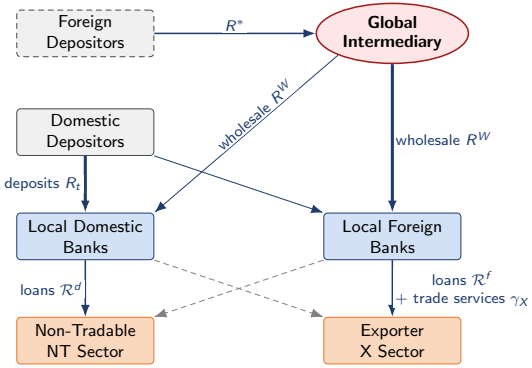




- Foreign banks rely more on cross-border and non-core funding, which result in lower funding costs
- Their credit supply reacts more to foreign shocks Morais et al. '19; Cantú et al. '22; Bruno and Shin '23

Model

Model Architecture Overview



A real, flexible-price SOE with:

- Two productive sectors $S \in \{NT, X\}$: non-traded and exporters, who need working-capital loans
- Two local banking sectors $b \in \{d, f\}$: domestic and foreign-owned banks; firms sort between them
- Foreign banks rely more on wholesale funding, $w^f > w^d$, and offer trade services to exporters γ_X

Financial transmission of a foreign interest rate tightening:

$$R^* \uparrow \Longrightarrow R^W \uparrow \Longrightarrow \underbrace{\mathcal{R}^d \uparrow,\quad \mathcal{R}^f \uparrow\uparrow}_{\text{loan rates}} \Longrightarrow \underbrace{y^d \downarrow,\quad y^f \downarrow\downarrow}_{\text{output of each bank's borrowers}}$$

Households

A representative household chooses consumption, hours and deposits to maximize

$$\mathbb{E}_0 \sum_{t \geq 0} \beta^t \frac{\widetilde{C}_t^{1-\sigma^h} - 1}{1-\sigma^h}, \quad \widetilde{C}_t = C_t - \frac{\zeta_0}{1+\zeta} H_t^{1+\zeta}, \quad \text{s.t.} \quad C_t + D_t = \mathrm{w}_t H_t + R_{t-1} D_{t-1} + \Pi_t.$$

- C_t is a nested CES of non-traded goods, home tradables and imports. The basket is the numeraire; imports cost q_t, the real exchange rate (rise = depreciation)
- Households save in home bank deposits: D_t deposited at t repays $R_t D_t$ at $t+1$
- Π_t includes home firms' profits and net payouts of domestic banks

▸ details

Firms: Technology

- A continuum of firms j in each sector $S \in \{NT, X\}$. Each firm is attached to one bank type $b \in \{d, f\}$, from which it borrows to pay for its inputs

Technology: $$y_{j,t}^{NT} = A_{NT,t}\, h_{j,t}, \qquad y_{j,t}^{X} = A_{X,t}^{b} \underbrace{m_{j,t}^{\alpha_X}}_{\text{imported inputs}} \underbrace{h_{j,t}^{1-\alpha_X}}_{\text{labor}}$$

$$A_{X,t}^{b} = \begin{cases} A_{X,t} & \text{if } b = d \\ \gamma_X A_{X,t} & \text{if } b = f \end{cases} \qquad \gamma_X \geq 1 : \text{better trade services at foreign banks}$$

Input cost: $$MC_{NT,t} = \frac{\mathrm{w}_t}{A_{NT,t}}, \qquad MC_{X,t}^{b} \propto \frac{\mathrm{w}_t^{1-\alpha_X}\, q_t^{\alpha_X}}{A_{X,t}^{b}} \implies MC_{X,t}^{f} = \frac{MC_{X,t}^{d}}{\gamma_X}$$

▸ foreign bank trade services

Firms: Working-Capital Finance and Pricing

- At t a firm borrows its input bill, produces and sells; at $t+1$ it repays
- NT firms borrow in pesos; exporters borrow a fraction $\alpha_{\$}$ in dollars

Loan rate: $\mathcal{R}^b_{NT,t} = R^b_{P,t}, \qquad \mathcal{R}^b_{X,t} = (1-\alpha_{\$}) R^b_{P,t} + \alpha_{\$}\, \mathbb{E}_t \frac{q_{t+1}}{q_t}\, R^b_{\$,t}$

Flexible prices: $p^b_{S,t} = \mu\, \mathcal{R}^b_{S,t}\, MC^b_{S,t}, \qquad \mu = \frac{\varepsilon}{\varepsilon - 1}$

- Loan rate enters as financing cost (Neumeyer-Perri '05, Akinci et al. '24)
- CES demand within each sector: a constant markup μ, and a sector price that averages the prices of firms at each bank
- Firms have no net worth and carry no risk. Banks are always repaid in full
- Individual firms have sticky bank relationships, and slowly re-sort across banks comparing costs, with idiosyncratic tastes ▸ details

Banks I: Balance Sheet and Set Up

Two home banking sectors $b \in \{d, f\}$ with atomistic banks. Their balance sheet:

$$\underbrace{L^b_{NT,t} + L^b_{X,t}}_{\substack{\text{loans} \\ \text{by sector}}} = \underbrace{L^b_t}_{\text{total loans}} = \underbrace{N^b_t}_{\text{net worth}} + \underbrace{D^b_t}_{\text{home deposits}} + \underbrace{W^b_t}_{\text{wholesale}} - \underbrace{\mathcal{C}^b_t}_{\substack{\text{wholesale} \\ \text{funding charge}}}$$

- N^b_{t+1} follows loan repayments minus repayments of deposits and wholesale details
- Bank exits with prob $1-\sigma$ and pays its net worth to owner, with SDF $\Lambda^b_{t,t+1}$:

$$V^b_t = \mathbb{E}_t\left[\Lambda^b_{t,t+1}\left((1-\sigma)N^b_{t+1} + \sigma V^b_{t+1}\right)\right]$$

- Banks can divert a fraction θ of loans, requiring $V^b_t \geq \theta L^b_t$ (Gertler-Kiyotaki)
- **The problem scales with net worth:** $V^b_t = \Psi^b_t N^b_t$, so the constraint limits lending to $L^b_t \leq \dfrac{\Psi^b_t}{\theta} N^b_t$

Banks II: The Wholesale Funding Charge

Raising W_t^b costs a charge paid upfront that depends on the wholesale ratio $w_t^b = \frac{W_t^b}{N_t^b}$:

$$C_t^b = \frac{\chi}{2}\left(w_t^b - \hat{w}^b\right)^2 N_t^b$$

- χ: how costly it is to move away from $\hat{w}^b$
- $\hat{w}^b$: the bank's target wholesale ratio, which it sustains at no charge
- Access to parent funding gives foreign banks a higher target:

$$\hat{w}^f > \hat{w}^d$$

Banks III: Currency of Funding and the NOP Regulation

- Wholesale funds from the global intermediary: $W_t^b = \underbrace{q_t W_{\$,t}^b}_{\text{dollars}} + \underbrace{W_{P,t}^b}_{\text{pesos}}$
- Dollar funds cost R_t^W; peso funds are dollars plus a hedge, at a UIP premium:

$$R_{P,t}^W = R_t^W \left(1 + \text{prem}_t\right) \mathbb{E}_t \frac{q_{t+1}}{q_t}, \qquad \text{prem}_t > 0$$

- Dollar funds are cheaper but expose banks to currency risk, which regulators cap

FX Net Open Position (NOP) Limit — Basel: capital charge on FX risk

A prudential cap on the net short dollar position:

$$\underbrace{q_t W_{\$,t}^b}_{\text{dollar wholesale borrowing}} - \underbrace{a_t^b L_t^b}_{\text{dollar loans}} \leq \Gamma N_t^b$$

where $a_t^b \equiv \alpha_{\$} L_{X,t}^b / L_t^b$ is the fraction of the bank's loans in dollars

Banks IV: The Banker's Problem and Its Solution

Per unit of net worth, the banker chooses its loans and its funding to solve

$$\Psi_t^b = \max\ \mathbb{E}_t\left[\Omega_{t,t+1}^b\, g_{t+1}^b\right], \qquad \Omega_{t,t+1}^b \equiv \Lambda_{t,t+1}^b\left(1-\sigma+\sigma\Psi_{t+1}^b\right)$$

subject to BS, IC and NOP limit, where $g_{t+1}^b = N_{t+1}^b/N_t^b$ is the return on net worth

Optimality conditions, with both constraints binding: ▸ full problem

Lending: $$L_t^b = \frac{\Psi_t^b}{\theta}\, N_t^b$$

Wholesale funding: $$w_t^b = \hat{w}^b + \tfrac{1}{\chi}\left(1 - R_{P,t}^W/R_t\right)$$

Dollar wholesale: $$q_t W_{\$,t}^b = a_t^b L_t^b + \Gamma N_t^b$$

Dollar loan rate: $$\underbrace{\mathbb{E}_t\left[\Omega_{t,t+1}^b\left(R_{P,t}^b - \frac{q_{t+1}}{q_t}R_{\$,t}^b\right)\right]}_{\text{extra return on a peso loan}} = \underbrace{\mathbb{E}_t\left[\Omega_{t,t+1}^b\left(R_{P,t}^W - \frac{q_{t+1}}{q_t}R_t^W\right)\right]}_{\text{extra cost of peso funding}}$$

The Rest of the World

To the home economy, the rest of the world supplies two things:

- **Wholesale funding**, lent at a spread z_t over the US rate R_t^*:

$$R_t^W = R_t^* (1 + z_t) \qquad R_{P,t}^W = R_t^W (1 + \text{prem}_t) \mathbb{E}_t \frac{q_{t+1}}{q_t}$$

- **Demand for exports:** $M_{C,t}^* = \omega_t^* C_t^* \left(\frac{p_{X,t}}{q_t} \right)^{-\eta^*}$

- A US monetary tightening moves R_t^*, z_t and C_t^* together: their joint process is fitted to VAR responses to an identified US monetary policy shock (Bauer-Swanson)
- The peso premium prem_t and export taste ω_t^* follow independent AR(1) processes

Equilibrium

Given the US block (R_t^*, z_t, C_t^*), the peso premium, export taste and productivity:

- Households choose consumption, hours and deposits
- Firms set flexible prices and slowly re-sort across banks
- Banks choose lending and funding, subject to the two constraints
- Markets for goods, labor, deposits and loans clear; the exchange rate q_t clears the balance of payments details

Solved to first order, around a steady state where both bank constraints bind

Transmission: Mechanism and Evidence

Loan Rates and Quantities

A firm in sector S at bank b faces CES demand and prices at $\mu \mathcal{R}_S^b MC_S$:

$$d \log y_S^b = \underbrace{d \log Y_S + \varepsilon \left(d \log p_S - d \log MC_S \right)}_{\text{same for every firm in the sector}} - \varepsilon \, d \log \mathcal{R}_S^b$$

$$\Longrightarrow \qquad d \log y_S^f - d \log y_S^d = -\varepsilon \left(d \log \mathcal{R}_S^f - d \log \mathcal{R}_S^d \right)$$

- Sector output falls with the sector's average loan rate $\mathcal{R}_S$, at demand elasticity η_S:

$$d \log Y_S = d \log \mathrm{dem}_S - \eta_S \left(d \log MC_S + d \log \mathcal{R}_S \right)$$

- dem_S: home consumption and, for exporters, US demand $\omega^* C^*$

A Rise in R^*: Which Banks Tighten Credit More?

- $R^* \uparrow$ raises wholesale rates; the deposit rate R follows as banks shift to deposits
- In response, loan rates rise with funding costs, weighted by each bank's mix:

$$dR_P^b = \frac{w^b \, dR_P^W + \delta^b \, dR}{\phi + \varepsilon} + \ldots \qquad w = \tfrac{W}{N},\ \delta = \tfrac{D}{N},\ \phi = \tfrac{L}{N}$$

▸ derivation

- With similar leverage, the foreign banks' extra wholesale replaces deposits:

$$dR_P^f - dR_P^d = \frac{w^f - w^d}{\phi + \varepsilon} \left(dR_P^W - dR \right) + \underbrace{\ldots}_{\text{loan demand, net worth, owners}}$$

▸ decomposition

- Foreign banks tighten more through funding if $dR_P^W > dR$, which requires $\chi > 0$
- The exchange rate enters through funding costs, net worth and loan demand

▸ details

From the Model to the Data: Credit Supply by Bank Type

Model prediction: after a US tightening, comparable firms at foreign banks pay higher loan rates and get less credit. In the model:

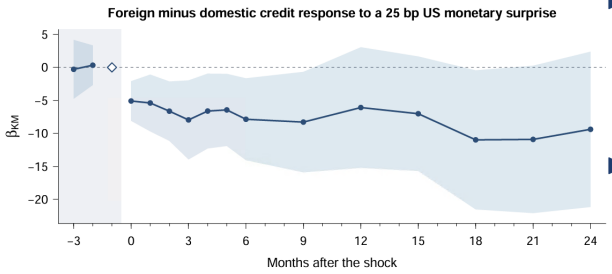
$$\beta_{KM} \equiv d\log L_S^f - d\log L_S^d = -\varepsilon\left(d\log \mathcal{R}_S^f - d\log \mathcal{R}_S^d\right)$$

Khwaja–Mian local projections. For each horizon h, compare credit growth across banks lending to *the same type of firm*:

$$g_{fb,t}^h = \alpha_b^h \;+\; \alpha_{d(f),t}^h \;+\; \beta_{KM}^h \left(\text{Foreign}_b \times \varepsilon_t^{MP}\right) \;+\; e_{fbt}^h$$

- $g_{fb,t}^h$: growth of firm–bank credit from the month before the shock to $t+h$, including entries and exits
- ε_t^{MP}: monthly US MP surprise, 2007–2019 (Bauer–Swanson)
- $\alpha_{d(f),t}^h$: industry $\times$ state $\times$ size $\times$ exporter $\times$ month (demand); α_b^h: bank

Foreign Banks Cut Credit by More After a US Tightening



- **Credit:** Foreign banks' loans to comparable firms contract. Holds in total and peso loans, stronger on extensive margin
- **Loan rates:** rise by a few basis points more at foreign banks; imprecisely estimated

- **Exports:** exporters attached to a foreign bank export less than comparable exporters at domestic banks
- Consistent with the evidence by Morais et al. '19 and Bruno-Shin '23

Quantitative Results

Calibration

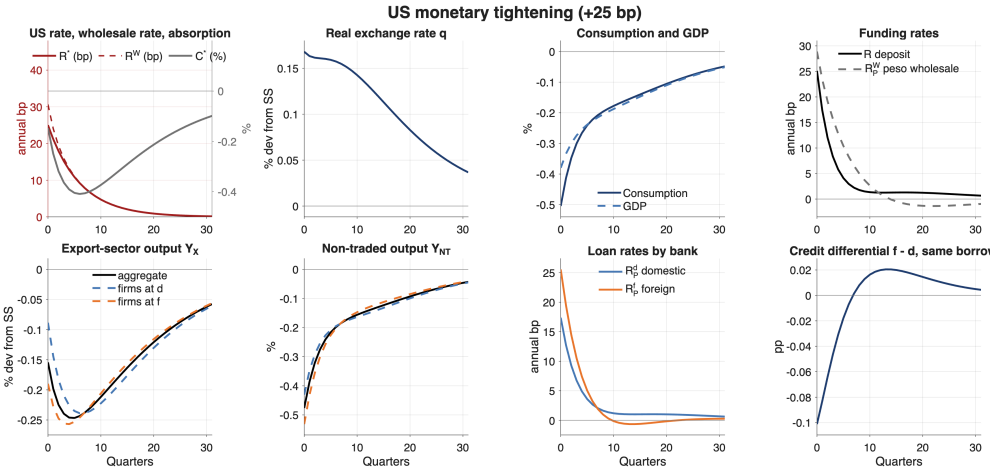
Parameter	Value	Targeted moment	Value	Source
$\hat{w}^d$, $\hat{w}^f$	1.11, 3.37	wholesale funding / equity, d and f	1.24, 3.51	CNBV
θ	0.268	bank leverage, risk-weighted	6.23	CNBV
ι^d, ι^f	.043, .044	peso loan rates, d and f (annual)	6.0%, 5.4%	registry
β	0.9896	banks' resident funding rate (annual)	4.2%	CNBV
ξ_{NT}, ξ_X	0.548, 0.353	domestic banks' credit share, NT and exporters	0.545, 0.35	registry
$\alpha_\$$	0.30	dollar share of exporters' credit	0.30	registry
$\bar{R}^*$, $\bar{z}$, $\overline{\text{prem}}$	1.0, 0.35, 2.0%	US real rate, dollar spread, peso premium (annual)		FRED, SIE
US block	fitted	responses of R^*, z, C^* to a US MP shock		Bauer–Swanson

Untargeted checks

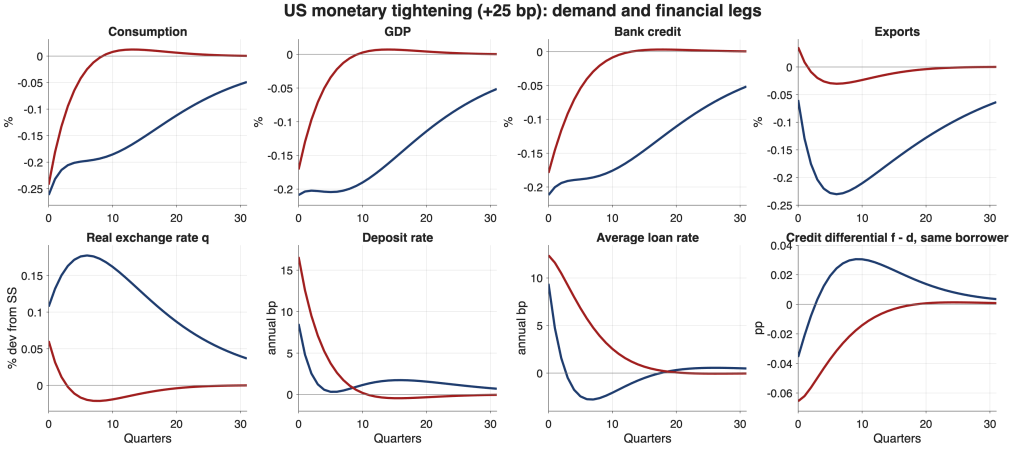
	Model	Data
Wholesale funding / equity, all banks	2.39	2.79
Foreign banks' share of business credit	.54	.63
Foreign banks' share of bank equity	.51	.65
Loan-rate gap $f - d$ bp within firm (\$/P)	-62	$-46/+92$
Responses to a US shock	▸ validation	

- **Trade services:** $\gamma_X = 1$; upper bound 1.05 ▸ evidence
- **Steady-state rates:** $\bar{R} = 4.2\% > \bar{R}_P^W = 3.3\% > \bar{R}^W = 1.4\% > \bar{R}^* = 1.0\%$
- **Externally set parameters** ▸ table

Foreign Interest Rate Shock u_{R^*}



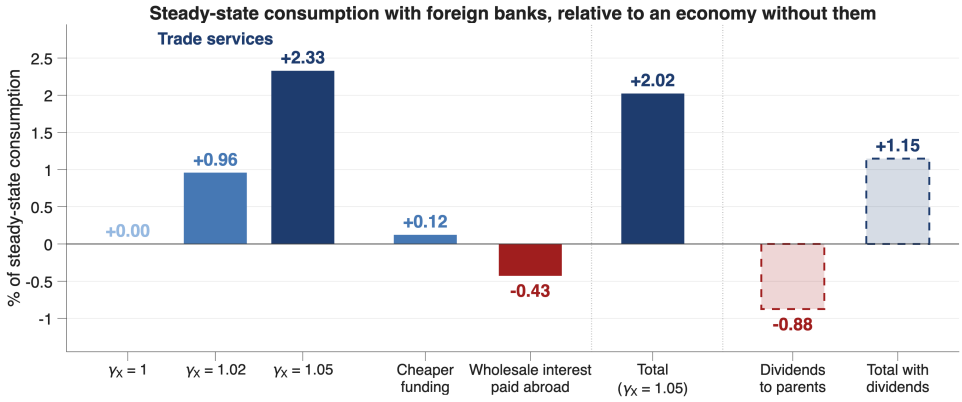
Foreign Interest Rate Decomposition



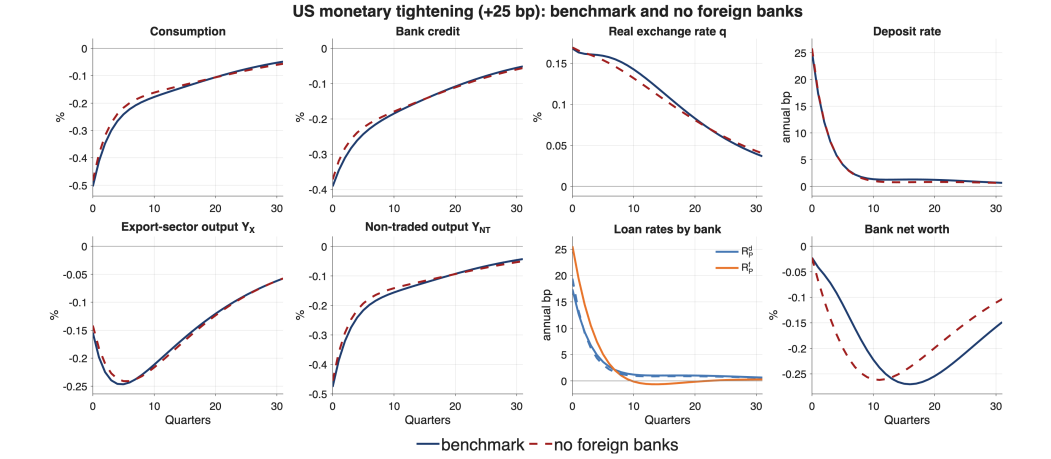
Other Shocks

- Peso premium financial shock ▸ IRF
- Global intermediary spread financial shock ▸ IRF
- Local productivity shock ▸ IRF
- Foreign demand shock ▸ IRF

No Foreign Bank Counterfactual: Steady State Decomposition

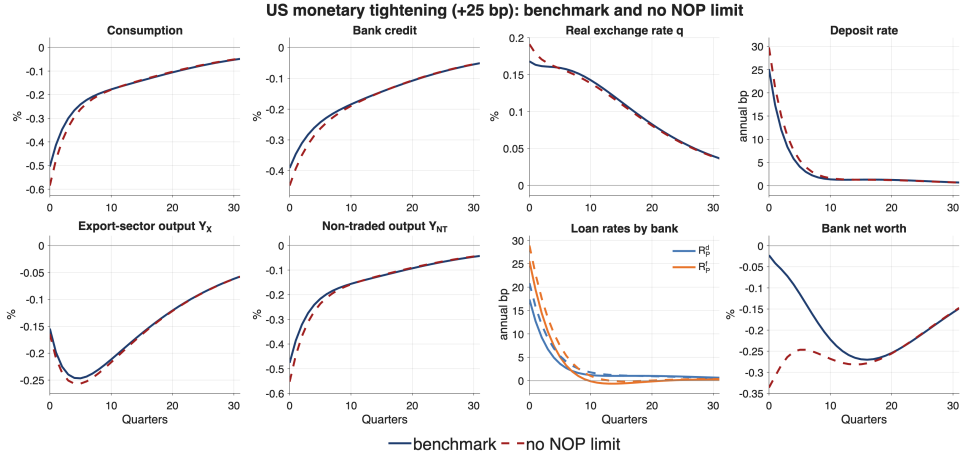


No Foreign Bank Counterfactual: Response to US Tightening



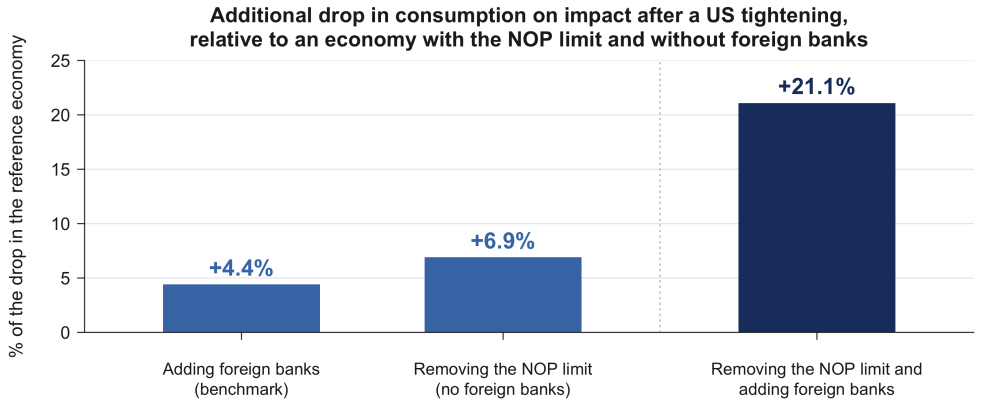
- The no foreign bank economy's response is only slightly muted – the economy is still exposed, borrowing from abroad

No NOP Regulation Counterfactual: Response to US Tightening



- Unregulated economy sees a larger response, particularly in bank net worth
- Steady-state consumption without the limit is 0.42% higher than in the benchmark, and the NT sector is relatively larger

Foreign Banks and the NOP Limit: Consumption Response



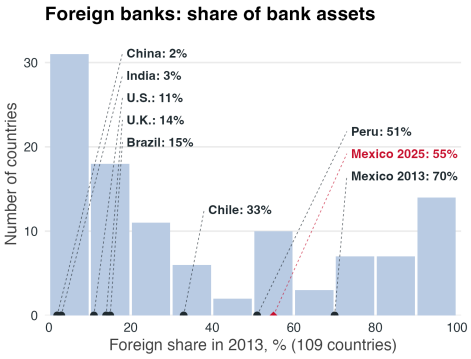
- The limit mitigates the additional response from introducing foreign banks table

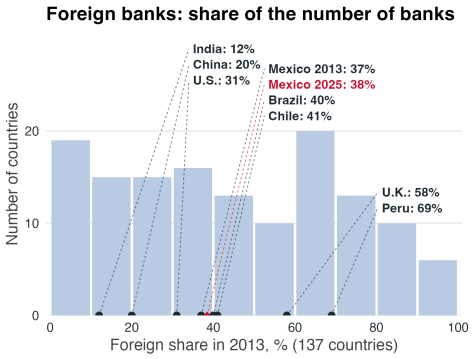
Conclusions

- Foreign banks lend and fund themselves differently from domestic banks
  - They tighten more than domestic banks after a US rate hike because their funding is priced off world rates
- Hosting them pays off mainly through trade services to exporters, which add up to 2% of steady state consumption
- The exposure they bring is modest: a US tightening hits consumption 4% harder on impact
- It is modest because regulation caps banks' dollar positions, which limits cross-border borrowing and net worth losses when the peso depreciates
  - Bank regulation mitigates the trade-off

Appendix

Foreign Bank Presence Across Countries

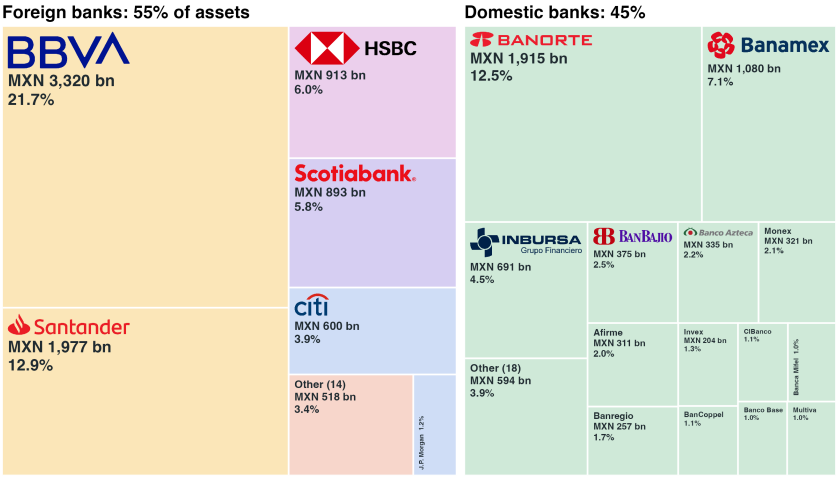




Sources: World Bank (Claessens and van Horen); CNBV.

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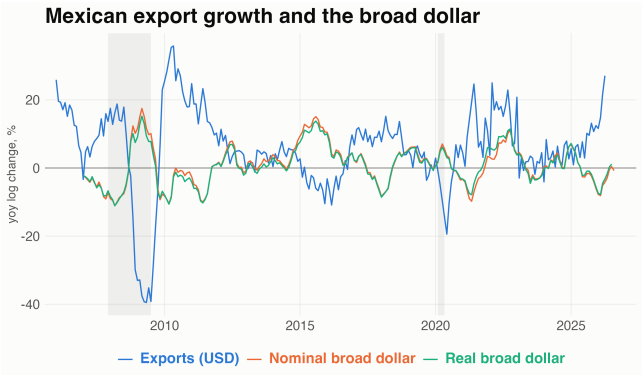
Mexican Banking System



Total assets of commercial banks, March 2025. Source: CNBV, Boletín Estadístico de Banca Múltiple.

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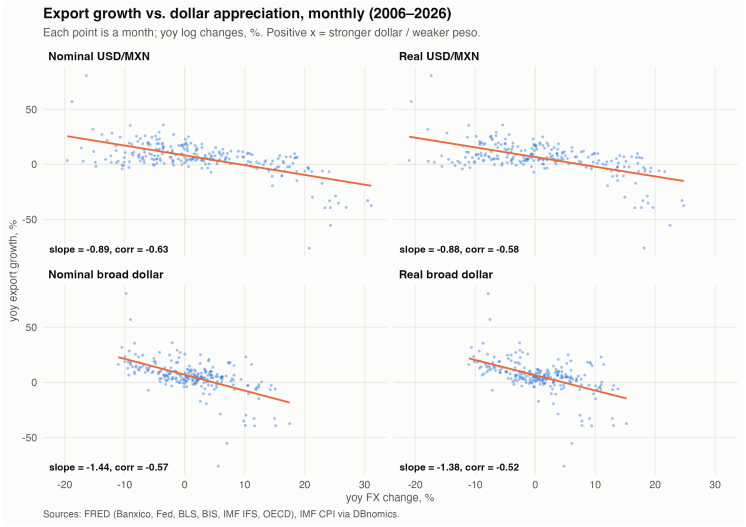
Exchange Rate and Exports



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- Mexican export growth is negatively correlated with dollar strength bilateral XR
- Counterintuitive from competitiveness channel
- But XR is endogenous, subject to both financial and demand forces!
  - Want info on how exporters are financed

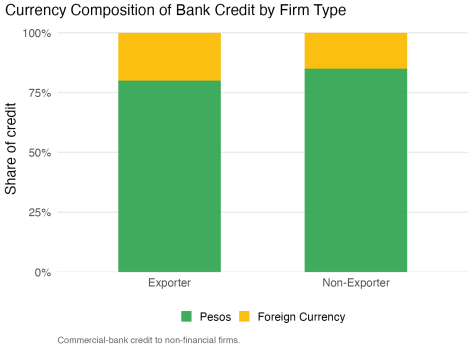
Negative Correlation Between Export Growth and Dollar Strength



back

Loan Pricing

	Interest Rate (%)	
	Peso	USD
log(Loan volume)	-0.752^{***}	-0.147^{***}
	(0.002)	(0.011)
international bank	0.918^{***}	-0.455^{***}
	(0.008)	(0.030)
Fixed Effects	Firm × Year	Firm × Year
Observations	7,102,149	72,005
Adjusted R^2	0.586	0.941



- Peso denominated loans are 90 bp more expensive when issued by an international bank, although dollar-denominated loans are cheaper
- But for both exporters and non-exporters most credit is in pesos

back

Dollar Appreciation and Foreign Banks' Funding Conditions

- Using bank-level survey we construct a measure of banks reported changes to factors affecting their credit supply (capitalization, liquidity, and availability of local- and foreign-currency funds)
- We regress this on the interaction of quarterly broad dollar growth and foreign ownership, controlling for demand with time fixed effects

	Mean funding conditions
Δ Broad dollar $\times$ Foreign	-0.99^{*}
	(0.56)
Fixed Effects	Bank + Time
Std. errors clustered by	Quarter
Residual df	1,599
R^2 (full / projected)	0.125 / 0.003

$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

$$\text{CreditSupplyFactors}_{b,t} = \beta\ (\Delta\%\,\text{Dollar}_t \times \text{Foreign}_b) + \alpha_b + \alpha_t + \varepsilon_{b,t}$$

back

Households

Consume and supply labor with GHH preferences with outer curvature σ^h:

$$\mathbb{E}_t \sum_{j\geq 0} \beta^j \frac{\widetilde{C}_{t+j}^{1-\sigma^h} - 1}{1-\sigma^h}, \qquad \widetilde{C}_t \equiv C_t - \frac{\zeta_0}{1+\zeta} H_t^{1+\zeta}, \qquad C_t + D_t = \mathrm{w}_t H_t + R_{t-1} D_{t-1} + \Pi_t,$$

where C_t is a nested CES composite:

$$C_t = \left[(1-\omega_T)^{\frac{1}{\eta}} C_{NT,t}^{\frac{\eta-1}{\eta}} + \omega_T^{\frac{1}{\eta}} C_{T,t}^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \qquad C_{T,t} = \left[(1-\omega_M)^{\frac{1}{\eta_T}} C_{X,t}^{\frac{\eta_T-1}{\eta_T}} + \omega_M^{\frac{1}{\eta_T}} M_{C,t}^{\frac{\eta_T-1}{\eta_T}}\right]^{\frac{\eta_T}{\eta_T-1}}$$

Optimality:

$$\beta\, \mathbb{E}_t\left[\left(\widetilde{C}_t / \widetilde{C}_{t+1}\right)^{\sigma^h}\right] R_t = 1, \qquad \mathrm{w}_t = \zeta_0 H_t^{\zeta},$$

$$C_{NT,t} = (1-\omega_T)\, p_{NT,t}^{-\eta} C_t, \qquad C_{T,t} = \omega_T\, p_{T,t}^{-\eta} C_t, \qquad M_{C,t} = \omega_M \left(\frac{q_t}{p_{T,t}}\right)^{-\eta_T} C_{T,t},$$

$$p_{T,t} = \left[(1-\omega_M)\, p_{X,t}^{1-\eta_T} + \omega_M\, q_t^{1-\eta_T}\right]^{\frac{1}{1-\eta_T}}.$$

back

Foreign Banks Market Trade Services

back





BBVA

Mejora tus actividades de exportación

Obtén los productos especializados para tu empresa exportadora y disminuye los riesgos latentes en el comercio internacional. Aborda los mercados extranjeros de manera eficiente con la confianza del apoyo de BBVA.

Banamex

Impuestos Aduanales

Permite gestionar de forma sencilla, rápida y segura, el pago de Impuestos Aduanales por importación y/o exportación de mercancía.

Banco Shinhan de México

Cartas de crédito

Este producto es ideal para importadores y exportadores que se dedican al comercio internacional, así como para empresas mayoristas o fabricantes locales.

Scotiabank

Comercio Exterior

Cartas de Crédito
- Domésticas
- Exportación

Cobranzas Documentarias
- Cobranza recibida
- Cobranza enviada

Financiamiento de Exportación
- Pago Anticipado de Exportación
- Pre-exportación

HSBC

Exportaciones

Adquiere un mayor control sobre tus exportaciones, mejora el flujo de efectivo y controla los riesgos relacionados con el comercio internacional de manera más efectiva con los servicios de financiamiento de exportaciones de HSBC.

Nuestras soluciones de exportaciones

Cartas de Crédito para la exportación

Realiza exportaciones con éxito al garantizar, sin el más mínimo riesgo y de manera confiable, la recepción de tu pago y la correcta entrega de la mercancía en el punto acordado.

Cobranza documentaria de Exportación

Asegura que la mercancía se libere sólo en el momento de recibir el pago o al tener la aceptación de una letra de cambio por parte del importador.

ICBC MÉXICO S.A.

- Financiamiento al comercio internacional
- Servicios al comercio internacional

Firms: Sorting Between Bank Types

- $\text{share}^b_{S,t}$: fraction of the firms in sector S attached to bank b
- Each period a firm keeps its bank with probability ς; otherwise it chooses again:

$$\text{share}^d_{S,t} = \underbrace{\varsigma \, \text{share}^d_{S,t-1}}_{\text{keep their bank}} + \underbrace{(1-\varsigma) \, \widehat{\text{share}}^{\,d}_{S,t}}_{\text{choose again}}$$

- Firms choosing again compare the current cost of producing with each bank, loan rate times input cost, subject to Gumbel taste shocks:

$$\widehat{\text{share}}^{\,d}_{S,t} = \frac{\xi_S \left(\mathcal{R}^d_{S,t} MC^d_{S,t}\right)^{-1/\nu}}{\xi_S \left(\mathcal{R}^d_{S,t} MC^d_{S,t}\right)^{-1/\nu} + (1-\xi_S) \left(\mathcal{R}^f_{S,t} MC^f_{S,t}\right)^{-1/\nu}}$$

- As if firms had CES demand for each bank: elasticity $1/\nu$, weights ξ_S
- Foreign banks attract exporters with a lower loan rate or with trade services ($MC^f_{X,t} = MC^d_{X,t}/\gamma_X$)

back

Net Worth

$$N^b_{t+1} = \underbrace{R^b_{P,t}(1-a^b_t)L^b_t + \frac{q_{t+1}}{q_t}R^b_{\$,t}\,a^b_t L^b_t}_{\text{loan repayments}} - \underbrace{R_t D^b_t}_{\text{deposits}} - \underbrace{q_{t+1}R^W_t\,W^b_{\$,t} - R^W_{P,t}W^b_{P,t}}_{\text{wholesale}} \equiv g^b_{t+1}N^b_t$$

- g^b_{t+1} is the gross return on net worth. Every rate is fixed when the contract is signed at t, so the only surprise at $t+1$ is the exchange rate
- Higher funding costs at t lower g^b_{t+1} only if loan rates at t rise by less: the margin earned on the whole book shrinks

Sector net worth: a fraction $1-\sigma$ of bankers exit each period and pay out their net worth to the owner; entrants bring in a fraction ι^b of last period's sector net worth:

$$N^b_t = \left(\sigma g^b_t + \iota^b\right) N^b_{t-1}$$

back

The Banker's Problem in Full, Per Unit of Net Worth

Choices: leverage $\phi_t = L_t/N_t$; dollar share of loans a_t; deposits $\delta_t = D_t/N_t$; wholesale $w_{\$,t} = q_t W_{\$,t}/N_t$, $w_{P,t} = W_{P,t}/N_t$, $w_t = w_{\$,t} + w_{P,t}$

Return: $g_{t+1} = \underbrace{R_{P,t}(1-a_t)\phi_t + \frac{q_{t+1}}{q_t} R_{\$,t}\, a_t \phi_t}_{\text{loan repayments}} - \underbrace{R_t \delta_t}_{\text{deposits}} - \underbrace{\frac{q_{t+1}}{q_t} R_t^W\, w_{\$,t} - R_{P,t}^W\, w_{P,t}}_{\text{wholesale}}$

Problem: $\Psi_t = \max_{\phi_t,\, a_t,\, \delta_t,\, w_{\$,t},\, w_{P,t}} \mathbb{E}_t\Big[\Lambda_{t,t+1}\big(1 - \sigma + \sigma\Psi_{t+1}\big)\, g_{t+1}\Big]$ subject to

$$\begin{aligned}
\text{balance sheet:} \quad & \phi_t = 1 + \delta_t + w_t - \tfrac{\chi}{2}(w_t - \hat{w})^2 \\
\text{incentive constraint:} \quad & \mathbb{E}_t\Big[\Lambda_{t,t+1}\big(1 - \sigma + \sigma\Psi_{t+1}\big)\, g_{t+1}\Big] \geq \theta\phi_t \\
\text{NOP limit:} \quad & w_{\$,t} - a_t\phi_t \leq \Gamma \\
\text{feasibility:} \quad & \delta_t,\ w_{\$,t},\ w_{P,t} \geq 0, \qquad 0 \leq a_t \leq \alpha_{\$}
\end{aligned}$$

$$\Lambda^d_{t,t+1} = \beta\Big(\frac{\widetilde{C}_t}{\widetilde{C}_{t+1}}\Big)^{\sigma^h}, \quad \Lambda^f_{t,t+1} = \frac{1}{R^*_t}\,\frac{q_t}{q_{t+1}}$$

- The banker takes rates, the exchange rate and Ψ_{t+1} as given. Net worth is the only state

back

Equilibrium, BOP, and the Steady State

Equilibrium: households, firms and banks optimize, firms sort, and all markets clear

XR q_t adjusts, jointly with trade and new borrowing, to clear BOP:

$$\underbrace{\sum_b W_t^b}_{\text{new wholesale borrowing}} = \underbrace{q_t R_{t-1}^W \sum_b W_{\$,t-1}^b + R_{P,t-1}^W \sum_b W_{P,t-1}^b}_{\text{repayment of last period's wholesale debt}} - \underbrace{TB_t}_{\text{trade balance}} - \underbrace{\mathcal{F}_t}_{\text{net equity inflow from parents}}$$

$$TB_t = \underbrace{p_{X,t} M_{C,t}^*}_{\text{exports}} - \underbrace{q_t \left(M_{C,t} + M_{X,t}\right)}_{\text{imports}} - \underbrace{\textstyle\sum_b \mathcal{C}_t^b}_{\text{funding charges}}$$

- **Steady state:** $\bar{q} = 1$, rates ordered $\bar{\mathcal{R}}_{NT}^b > \bar{\mathcal{R}}_X^b > \bar{R} > \bar{R}_P^W > \bar{R}^W > \bar{R}^*$
- **Solution:** first order, around steady state where the IC and NOP limit bind. Responses stay in this regime, but can extend to occasionally-binding

back

From Funding Costs to Loan Rates

Cost of funds $\mathfrak{c}$. What the bank pays its three claimants, weighted by their size relative to net worth, $\delta = \frac{D}{N}$, $w = \frac{W}{N}$ and 1, over leverage $\phi = \frac{L}{N}$:

$$d\mathfrak{c} \approx \underbrace{\tfrac{\delta}{\phi}\, dR}_{\text{deposits}} + \underbrace{\tfrac{w}{\phi}\, dR_P^W}_{\text{wholesale}} + \underbrace{\tfrac{1}{\phi}\, d\log R_O}_{\text{owner}}, \qquad R_O = \begin{cases} R & \text{domestic} \\ R^* \,\mathbb{E}\, q'/q & \text{foreign} \end{cases}$$

Loan rate. The margin $R_P - \mathfrak{c}$ is earned on all ϕ units of loans per unit of net worth. At given loan demand and net worth:

$$\text{supply: } d\log L = \phi\,(dR_P - d\mathfrak{c}) \qquad \text{demand: } d\log L = -\varepsilon\, dR_P$$

▸ details

$$\Longrightarrow \qquad \frac{\partial R_P}{\partial \mathfrak{c}} = \frac{\phi}{\phi+\varepsilon}, \qquad \frac{\partial R_P}{\partial R_P^W} = \frac{w}{\phi+\varepsilon}$$

- The wholesale rate passes through more at the foreign bank: $w^f > w^d$

◂ back

Demand for a Bank's Loans

A firm in sector S at bank b borrows its input bill, $MC_S\, y_S^b$. From its output equation,

$$d\log\left(MC_S\, y_S^b\right) = \underbrace{d\log Y_S + \varepsilon\, d\log p_S - (\varepsilon-1)\, d\log MC_S}_{\text{same for every firm in the sector}} - \varepsilon\, d\log \mathcal{R}_S^b$$

Adding up over the bank's borrowers, with $\bar{L}_S^b/\bar{L}^b$ the share of sector S in its loans:

$$d\log L^b = d\log \mathcal{D}^b - \varepsilon\, dR_P^b$$

$$d\log \mathcal{D}^b = \sum_S \frac{\bar{L}_S^b}{\bar{L}^b}\Big[\underbrace{d\log Y_S + \varepsilon\, d\log p_S - (\varepsilon-1)\, d\log MC_S}_{\text{sector demand, prices and input costs}} + \underbrace{d\log \text{share}_S^b}_{\text{firms attached}}\Big]$$

- Demand for the two banks' loans differs through the sector mix of their borrowers
- A fall in export demand lowers $\mathcal{D}^b$ more at the bank that lends more to exporters

back

The Loan-Rate Gap: Full Decomposition

Solving the gap forward, with similar leverage $\bar{\phi}$ and $\kappa = \dfrac{\varepsilon}{\bar{\phi}+\varepsilon} < 1$:

$$dR^f_{P,t} - dR^d_{P,t} = \frac{1}{\bar{\phi}+\varepsilon} \sum_{j\geq 0} \kappa^j \, \mathbb{E}_t \Big[\underbrace{(\bar{w}^f - \bar{w}^d)\big(dR^W_P - dR\big)_{t+j}}_{\text{funding}} + \underbrace{\bar{g}\, d\log \frac{R^f_{O,t+j}}{R_{t+j}}}_{\text{owners}} + \underbrace{x_{t+j} - x_{t+j+1}}_{\substack{\text{loan demand} \\ \text{and net worth}}} \Big]$$

where $x_t \equiv d\log \dfrac{\mathcal{D}^f_t}{\mathcal{D}^d_t} - d\log \dfrac{N^f_t}{N^d_t}$: relative loan demand minus relative net worth

- **Funding** counts in every period: a wholesale rate above the deposit rate later lowers the foreign bank's value and its lending today
- **Loan demand and net worth:** strong demand or scarce net worth later raises a bank's value and lets it lend more now

back

Exchange Rate Channels

Output and loan demand: the level of q

- Imported inputs: $d \log MC_X = (1 - \alpha_X)\, d \log \mathrm{w} + \alpha_X\, d \log q$
- Foreign demand for exports rises with q (comp channel), inside dem_X and $\mathcal{D}^b$

Funding costs: expected depreciation raises R_P^W for both banks and R_O^f for the foreign owner. Direct effect on the loan rate:

$$\frac{\partial R_P^d}{\partial \mathbb{E} \log(q'/q)} = \frac{\bar{w}^d}{\bar{\phi}^d + \varepsilon}, \qquad \frac{\partial R_P^f}{\partial \mathbb{E} \log(q'/q)} = \frac{\bar{w}^f + \bar{g}}{\bar{\phi}^f + \varepsilon}$$

Net worth: a surprise depreciation lowers N^b through the short dollar position:

$$\frac{\partial \log N_t^b}{\partial \log q_t} = -\sigma \Big[\underbrace{\Gamma\, \bar{R}^W}_{\substack{\text{dollar debt above} \\ \text{dollar loans}}} - \underbrace{\bar{a}^b \bar{\phi}^b \left(\bar{R}_\$^b - \bar{R}^W\right)}_{\substack{\text{margin earned} \\ \text{on dollar loans}}} \Big]$$

- The NOP limit caps this loss per unit of depreciation

back

Calibration: Externally Set Parameters

		Value	Source
σ^h	household curvature	0.5	rate responses
ζ	inverse Frisch elasticity	0.306	Leyva–Urrutia (2020)
ε	variety elasticity	5	markup 1.25
$\eta,\ \eta_T,\ \eta^*$	NT–tradable, home–import, export elasticities	0.5, 1.5, 1.5	Akinci (2011); standard
$\omega_T,\ \omega_M$	tradable and import weights	0.25, 0.45	exports/GDP ≈ 0.4
α_X	imported-input share of exporters	0.50	import content of exports
σ	banker survival	0.92	Aoki et al. (2021)
Γ	NOP limit, share of capital	0.15	Banco de México
ς	borrower retention	0.95	credit registry
ν	taste dispersion in bank choice	0.30	interest rate shock
χ	funding curvature, both banks	0.016	funding responses
γ_X	trade services	1	upper bound 1.05
Premium	level AR(1), $\rho = 0.02$, s.d. 3.5 pp		survey, 2015–19
Spread z	log AR(1), $\rho = 0.50$		fitted US block
$\omega^*,\ A$	AR(1), $\rho = 0.9$		not calibrated
$\zeta_0,\ \bar{C}^*$	normalizations, $\bar{H} = 1,\ \bar{q} = 1$	0.685, 0.486	

back

Do Specialized Foreign Banks Help Firms Export?

event study back

Clients of **specialized** foreign banks export far more (1.34 log points) than similar firms: is that just sorting, or does it hold within firm?

$$\ln \text{Exports}_{it} = \beta\, \text{SFB}_{i,t-1} + \delta \ln \text{Credit}_{i,t-1} + \phi\, \text{Foreign}_{i,t-1} + \alpha_i + \alpha_{s\times r\times t} + \varepsilon_{it}$$

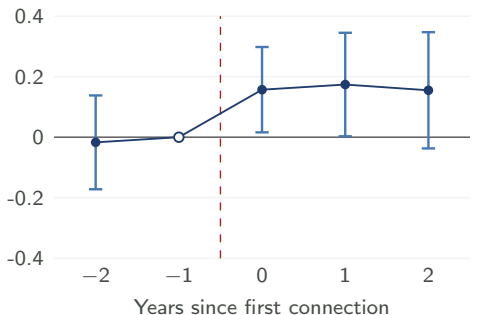
	Log exports$_t$		
	(1)	(2)	(3)
Spec foreign bank$_{t-1}$ (β)	0.199	0.177	0.176
	(0.069)	(0.069)	(0.069)
Log credit$_{t-1}$		✓	✓
Any foreign bank$_{t-1}$			✓
Firm FE	✓	✓	✓
Sector×state×year FE	✓	✓	✓
Observations		88,321	

- $\text{SFB}_{i,t-1}$: binary, credit from a specialized foreign bank at $t-1$
- $\text{Foreign}_{i,t-1}$: binary, credit from any foreign bank
- $\text{Credit}_{i,t-1}$: total credit received
- Exports are $\approx$ 19–22% higher in connected years $\Rightarrow$ Most of the cross-firm premium is sorting but not all

Export years 2015–2023; positive-export years; $s \times r \times t$ = sector × state × year; firm-clustered SEs.

Event Study and Mapping to γ_X

Same regression with event-time dummies around a firm's *first* connection to a specialized foreign bank, reference year -1:



Log exports; controls as before. 516 switchers, 24,412 never-connected controls, $N = 86{,}935$. Bars: 90% CI.

From β to γ_X. At fixed demand, revenue moves with cost at elasticity $\varepsilon - 1$:

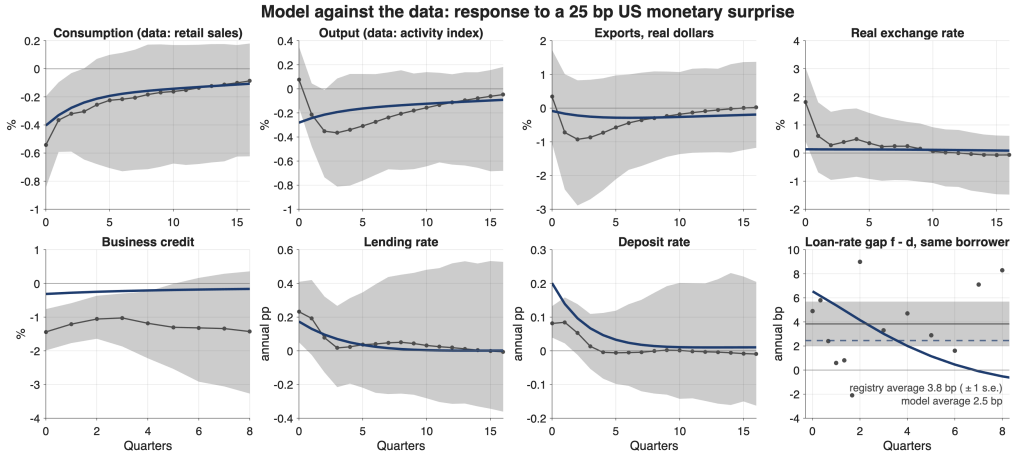
$$\Delta \ln(py) = (\varepsilon - 1)\left[\ln \gamma_X - \Delta \ln \mathcal{R}_X\right]$$

1. Foreign peso loans are more expensive, so assigning all of β to services is a lower bound: $\ln \gamma^{spec} \geq \beta/(\varepsilon - 1) \approx 0.05$
2. Specialized banks hold $\approx 5\%$ of exporters' credit, and the big five show no premium $\Rightarrow$ benchmark $\gamma_X = 1$
3. Upper bound $\gamma_X = 1.05$: every foreign-bank exporter gets the specialized premium

◂ back ▸ regression

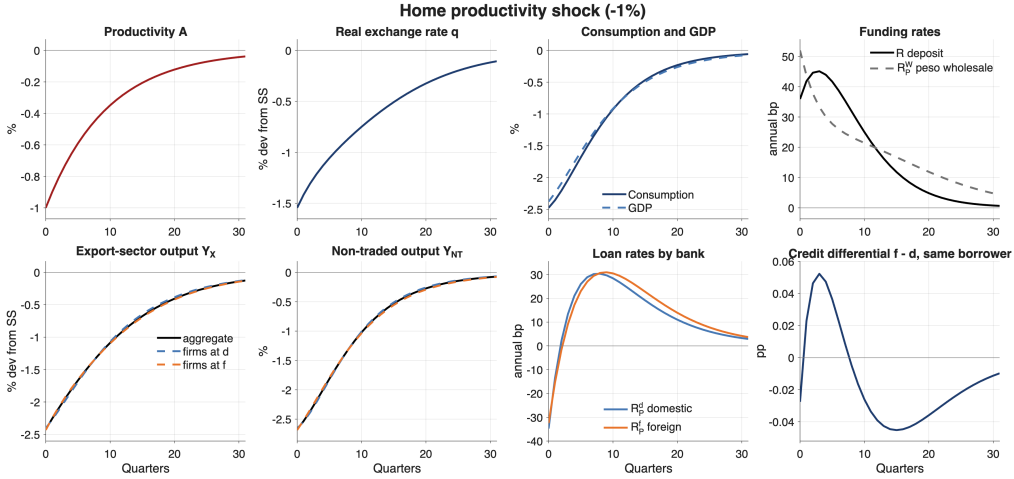
Model validation vs VAR

back



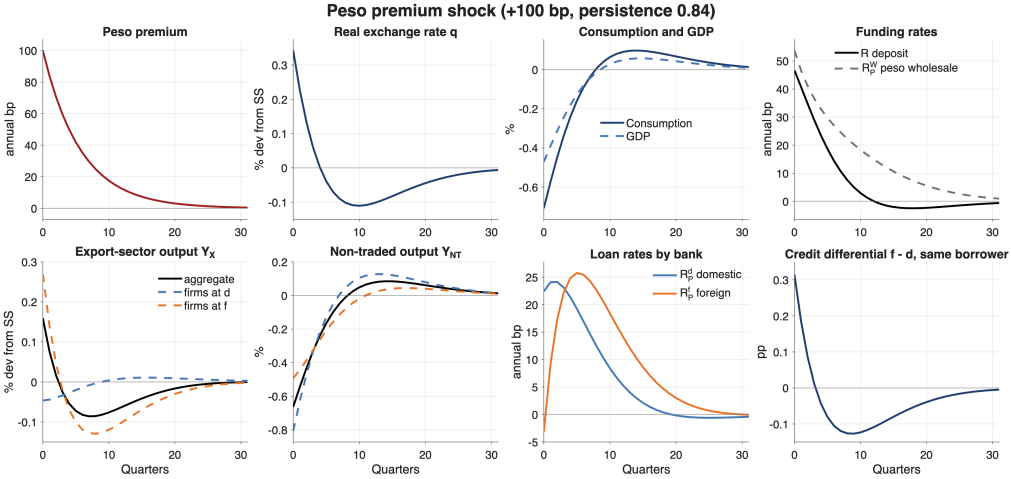
Local Productivity Shock u_A

back



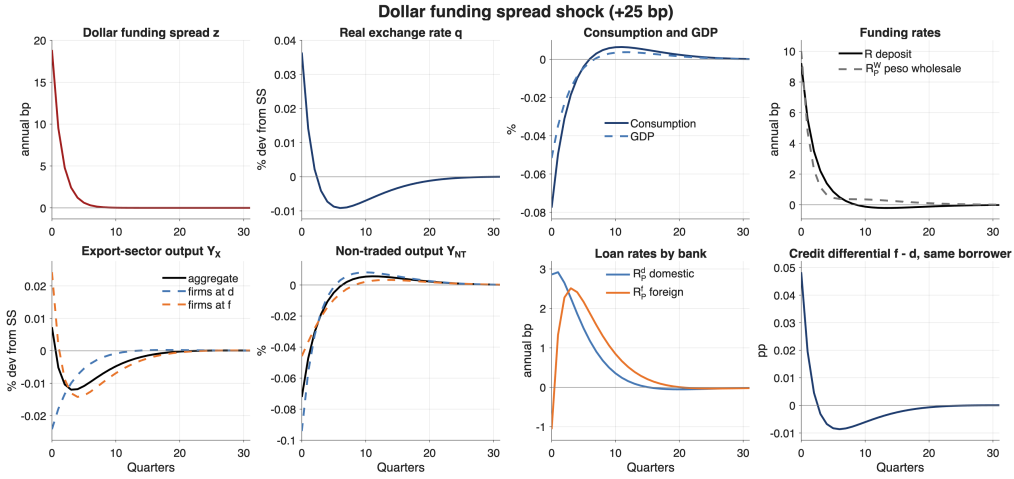
Peso Premium Shock

back



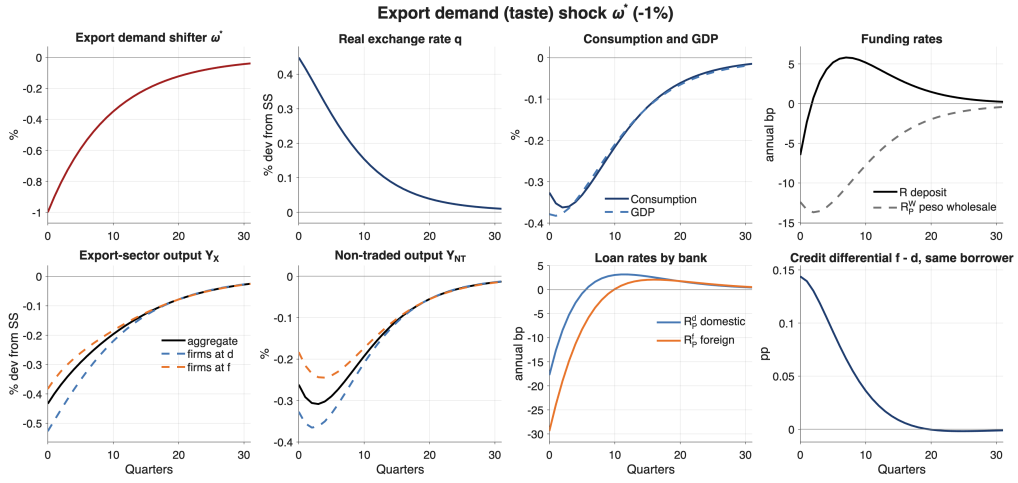
Dollar Spread Shock u_z

back



A Foreign Demand Shock

back



Foreign Banks and the NOP Limit: Consumption Response

back

Consumption response (%) to a 25 bp US tightening: impact / cumulative over two years

	Both banks	**No foreign banks**
NOP limit	-0.50 / -2.50	-0.48 / -2.30
No limit	-0.59 / -2.80	-0.52 / -2.45